

Cadence

Software segment — presentation script

4-speaker group talk ■ your part ■ target $\approx$ 3 minutes

Before you start

- Let *Hardware* hand you “a stream of samples” — *you* own the Freeze Index, gate & debounce.
 - Frame your pipeline as the *hand-built, rule-based* detector (it sets up the ML handoff).
 - Running long? The “Host tooling” slide is the safe cut. Bold words = stress them.
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SLIDE ■ from raw motion to a live decision

0:00

Thanks. So [Speaker 2] hands the software a stream of accelerometer samples at **64 hertz**. My job is everything that turns that raw stream into a freeze decision you can **trust** — both on the device in real time, and on the host afterwards.

SLIDE ■ on-device pipeline

0:25

Here's what runs on the board itself. Every 2 seconds, I take the last 4-second window — 256 samples — and run a short **FFT** over it. I sum the power in the freeze band, 3 to 8 hertz, and the power in the walking band, half to 3 hertz, and divide them — that ratio is the **Freeze Index**.

But — and this is the important part — a high index on its own is *not* a freeze. The decision logic only says *freeze* when **two conditions** are met: there's enough overall movement energy to clear the **gate**, *and* the index is over **threshold**. Then a **two-window debounce** state machine requires that to hold for two readings in a row before it flips the state, on or off. That single piece of logic is what kills the one-off false alarms. The result drives the on-board display live — and yes, the cue path exists in the code, but it's **disabled in firmware** by design.

And all of that — an FFT and a decision, twice a second — runs in

lightweight integer math on a tiny low-power microcontroller. No phone, no cloud.

SLIDE ■ host tooling & data pipeline

1:40

Off the device, the software does two more jobs. The firmware tags **every sample with an auto-incrementing phase label** and logs it to flash; we replay it over USB straight into a CSV, so we get **ground truth for free**. And a Python pipeline windows that data, builds the confusion matrices and the threshold sweep — that's what let us tune the threshold **honestly**, not by eye. The same decision stream also feeds a **live web dashboard** — which I'll show you right now.

SLIDE ■ live demo

2:10

(start the dashboard replay) This is a real capture playing back. Watch the Freeze Index trace against the operating line. Through normal walking it stays **under the line** — silent. And here, the genuine freeze — it jumps above, the state flips to **FREEZE**, the dashboard lights up. Quiet on the walk, clean catch on the freeze — that's the gate and the debounce, in software, doing exactly their job.

SLIDE ■ close

2:50

So the whole pipeline is deliberately **transparent** — rule-based, traceable, every decision explainable. That's the philosophy behind Cadence: *a glass-box monitor a clinician can trust beats a black box that just buzzes*. Thank you — happy to take questions.